ATIF ASSISTANT - FULL EXPANDED PLAN

Beyond Jarvis • Local-First • Human+AI Hybrid

Planning Document - All Questions Addressed

EXECUTIVE SUMMARY

Atif Assistant is a privacy-first, local-first personal intelligence system. It reads your real data, learns context per person, extracts actionable tasks automatically, and builds spatial understanding via affordable glasses. Every phase is designed to work standalone.

CORE PRINCIPLES

- Local-first & offline-capable (no mandatory cloud)
- Read-only first, propose-then-approve for all connectors
- Phase independence (missing phase won't break system)
- Evidence-based labels (FACT/INFERENCE/ASSUMPTION/UNKNOWN/PREDICTION)
- Advisory-only (speaker ID never stored by itself)
- Human+AI hybrid with measured emotional awareness
- Incremental, piecewise reconstruction of incomplete data
- Pragmatic & affordable

ALL QUESTIONS ADDRESSED

Q1 Vision: human+AI hybrid, local-first, emotions

Q2 Connectors: Drive/Gmail/WhatsApp, read+update, memory sync

Q3 Glasses: affordable shortlist, voice/tone/face, spatial

Q4 Task extraction: auto-detect intents, todo management

Q5 Data mgmt: SQLite+FTS+local embeddings, tiering, no cloud

Q6 Offline LLM: how they work, limits, piecewise learning

Q7 Phased: easy->hard, each standalone

Q8 Human understanding + emotions

Q9 Piecewise merge of incomplete info

Q10 Archive all (done)

Q11 Spatial/home 3D mapping

Q12 Voice/tone/face recognition per person

Q13 Capability estimation (strength vs object)

Q14 Flowcharts/tree/visuals/tools

DETAILED PHASES (EASY TO HARD)

P0 - Foundation

Stabilize atif-assistant core; CRUD complete; 392/0 tests; preserve isolation

P1 - Connectors (Read-Only)

Drive/Gmail/WhatsApp/local. Scan -> propose -> approve. Dedup, provenance, evidence

P2 - Task Extraction

Detect 'I will...', dates/times, entities; auto-add to todos; link to people/episodes

P3 - Memory Enrichment

Auto-update facts from data; link people/locations; prune irrelevant

P4 - Perception (Glasses)

On-device voice/tone/face; enroll with consent; advisory-only ID

P5 - Spatial & Home Mapping

Locations, objects, shapes/materials; build scene graph; home structure

P6 - Reasoning & Capability

Human+context reasoning; strength vs object feasibility; optimal methods

P7 - Offline LLM Optimization

Small models (Ollama), RAG local, chunking, piecewise merge, incremental learning

P8 - 3D Reconstruction

Generate home 3D from observations; progressive refinement

P9 - Polish & Local Deploy

Self-contained, no cloud; affordable, maintainable

AFFORDABLE HARDWARE - GLASSES

Raspberry Pi Zero 2W + camera module - $35-50

Full local processing, WiFi, flexible, low power

ESP32-S3-EYE/ESP32-CAM - $15-30

Very cheap; streams to host; low compute

Open-source frames - $20-40

DIY mount; 3D-printable

Repurpose basic smart glasses - $40-70 (used)

Fastest path to test

DATA MGMT - HUGE DATA WITHOUT CLOUD

Smart approach: SQLite+FTS5, small local embeddings, hot/warm/cold tiering, compression, dedup, relevance-based pruning. Index what matters now, archive older evidence. Keep local-only.

- Tiering: hot (recent) in DB, cold (archive) compressed
- FTS5 for search, small embeddings for similarity
- Incremental indexing, dedup by content hash
- Prune by age/relevance; user controls

SYSTEM FLOW (HIGH LEVEL)

Input → Read-only Ingest (propose-then-approve) → Normalize → Evidence Store → Context Engine → Reasoning (human+AI) → Task Extractor → Memory Update → Action/Response. All local.

TREE STRUCTURE

Atif Assistant Core (evidence-backed)

■■ Connectors (Drive/Gmail/WhatsApp/Local)

■■ Memory (Facts/Episodes/People/Places/Tasks/Rules)

■■ Perception (Voice/Tone/Face/Scene/Spatial)

■■ Reasoning (Piecewise merge, Capability)

■■ Task Engine (Auto-extract + Todos)

■■ Local LLM (Ollama, small models)

This is planning-only. Built on existing atif-assistant foundation.